Real-Time Hub-to-Hub ETA Prediction with Kafka and Spark Structured Streaming

HyperLogistics-Stream • COMP-SCI 5540 Principles of Big Data Management, Fall 2026, UMKC • Instructor: Dr. Md Yusuf Sarwar Uddin

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Motivation

Middle-mile shipments run on fixed schedules, but a destination hub often learns how late an inbound trip is only on arrival, which limits dock-door and labor planning. Weather is a major driver: the FAA attributes 74.26% of system-impacting delays over 15 minutes to weather [9]. We build a streaming pipeline that predicts arrival delay *at departure time* from weather and lane history, and measure how fast and how well it does so.

Objectives and Goals

- **RQ1 (accuracy).** Do departure delay, streaming weather and lane history beat the "delay unchanged" baseline and the historical lane mean (MAE, RMSE)? Models: Spark MLlib Gradient-Boosted Trees and Random Forest, trained on 2018–2023 (2020 excluded), tested on 2024, with features known at departure only. Predicted ETA = scheduled arrival + departure delay + predicted change.
- **RQ2 (system performance).** How do cores (2/4/8), shuffle partitions (8/64/200), Kafka replay rate (1k/5k/20k events/s) and `maxOffsetsPerTrigger` (5k/20k/100k) change throughput, batch time and p50/p95 end-to-end latency [2, 8]?
- **RQ3 (stretch).** Can an LLM agent answer dispatcher questions from the live prediction tables, kept off the streaming path?

Data Sources

- **BTS Reporting Carrier On-Time Performance, 2018–2024**: tens of millions of flights with scheduled and actual times, delays and delay causes; raw CSV expected in the 5–10 GB range (to be confirmed after download). https://www.transtats.bts.gov/DL_SelectFields.aspx?gnoyr_VQ=FGJ&QO_fu146_anzr=b0-gvzr
- **NOAA Integrated Surface Database** [6]: hourly temperature, visibility and wind; free bulk access. https://www.ncei.noaa.gov/data/global-hourly/
- **Fusion:** each airport is mapped to its nearest NOAA station (ICAO match), local times are converted to UTC, and Spark joins on (station, UTC hour). BTS's weather-delay field is excluded, since it is known only after arrival and covers extreme weather only.

Proxy and limits. Real middle-mile telematics are proprietary, so scheduled hub-to-hub flights are a public proxy with the same structure (fixed lane, schedule, weather, destination congestion). BTS covers passenger carriers, not cargo carriers. Trucks add Hours of Service limits, road traffic and dwell; the stream replays recorded data. Related work: flight delays [4, 5], logistics ETA [7]; we found no study validating flights as a trucking proxy.

Architecture, Tools and Plan

Ingestion: Kafka producer replaying flights in WheelsOff order [3]. **Storage:** Parquet medallion (bronze raw events, silver cleaned and weather-enriched, gold features, predictions and sliding-window hub arrival counts). **Processing:** PySpark Structured Streaming [1] with stream-static joins and MLlib scoring. **Serving:** FastAPI and Streamlit dashboard, packaged with Docker Compose (one-command run). Optional stateful feature: previous-leg delay for the same tail number.
Schedule (8 weeks): W1 setup; W1–2 data and join; W2–4 features and models; W2–5 producer and streaming job; W4–6 API and dashboard; W5–7 benchmarks; W7–8 report and demo. If compute is tight, Random Forest tuning is cut first.

References

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